

Data Validations - Phase 2

```
In [16]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

print("Environment ready")
```

Environment ready

```
In [17]: # Load Excel Data

file_path = "Fantasy-Lord-Rookie-ML-Dataset.xlsx"

draft_df = pd.read_excel(file_path, sheet_name="draft_and_player_data")
fantasy_df = pd.read_excel(file_path, sheet_name="fantasy_targets")
master_df = pd.read_excel(file_path, sheet_name="master_dataset")

print("Draft Data:", draft_df.shape)
print("Fantasy Targets:", fantasy_df.shape)
print("Master Dataset:", master_df.shape)
```

Draft Data: (164, 14)
Fantasy Targets: (164, 7)
Master Dataset: (164, 17)

```
In [18]: # Preview Draft Table

draft_df.head()
```

Out[18]:

	player_id	player_name	season	position	nfl_team	draft_round	draft_pick_overall	age_on_draft_day	height_inches
0	qb_2010_bradford	Sam Bradford	2010	QB	Rams	1	1	22	76
1	rb_2010_spiller	C.J. Spiller	2010	RB	Bills	1	9	23	71
2	rb_2010_mathews	Ryan Mathews	2010	RB	Chargers	1	12	22	72
3	te_2010_gresham	Jermaine Gresham	2010	TE	Bengals	1	21	22	77
4	wr_2010_thomas	Demaryius Thomas	2010	WR	Broncos	1	22	22	75



In [19]:

Preview Fantasy Target Table

fantasy_df.head()

Out[19]:

	player_id	player_name	season	position	rookie_total_ppr_points	season_positional_finish	fantasy_relevant_binary
0	qb_2010_bradford	Sam Bradford	2010	QB	245.7	16.0	0.0
1	rb_2010_spiller	C.J. Spiller	2010	RB	95.2	49.0	0.0
2	rb_2010_mathews	Ryan Mathews	2010	RB	158.3	32.0	0.0
3	te_2010_gresham	Jermaine Gresham	2010	TE	112.1	16.0	0.0
4	wr_2010_thomas	Demaryius Thomas	2010	WR	52.3	87.0	0.0

In [20]:

Preview Master Table

master_df.head()

Out[20]:

	player_id	player_name	season	position	nfl_team	draft_round	draft_pick_overall	age_on_draft_day	height_inches
0	qb_2010_bradford	Sam Bradford	2010	QB	Rams	1	1	22	76
1	rb_2010_spiller	C.J. Spiller	2010	RB	Bills	1	9	23	71
2	rb_2010_mathews	Ryan Mathews	2010	RB	Chargers	1	12	22	72
3	te_2010_gresham	Jermaine Gresham	2010	TE	Bengals	1	21	22	77
4	wr_2010_thomas	Demaryius Thomas	2010	WR	Broncos	1	22	22	75

In [21]: *# Check Column Names*

```

print("Draft Columns:")
print(draft_df.columns.tolist())

print("\nFantasy Columns:")
print(fantasy_df.columns.tolist())

print("\nMaster Columns:")
print(master_df.columns.tolist())

```

Draft Columns:

```
['player_id', 'player_name', 'season', 'position', 'nfl_team', 'draft_round', 'draft_pick_overall', 'age_on_draft_day', 'height_inches', 'weight_lbs', 'college', 'college_games_played', 'career_yards', 'career_touchdowns']
```

Fantasy Columns:

```
['player_id', 'player_name', 'season', 'position', 'rookie_total_ppr_points', 'season_positional_finish', 'fantasy_relevant_binary']
```

Master Columns:

```
['player_id', 'player_name', 'season', 'position', 'nfl_team', 'draft_round', 'draft_pick_overall', 'age_on_draft_day', 'height_inches', 'weight_lbs', 'college', 'college_games_played', 'career_yards', 'career_touchdowns', 'rookie_total_ppr_points', 'season_positional_finish', 'fantasy_relevant_binary']
```

```
In [22]: # Check Missing Values

print("Draft Missing Values:")
display(draft_df.isnull().sum())

print("Fantasy Missing Values:")
display(fantasy_df.isnull().sum())

print("Master Missing Values:")
display(master_df.isnull().sum())
```

Draft Missing Values:

player_id	0
player_name	0
season	0
position	0
nfl_team	0
draft_round	0
draft_pick_overall	0
age_on_draft_day	0
height_inches	0
weight_lbs	0
college	0
college_games_played	0
career_yards	0
career_touchdowns	0

dtype: int64

Fantasy Missing Values:

player_id	0
player_name	0
season	0
position	0
rookie_total_ppr_points	10
season_positional_finish	10
fantasy_relevant_binary	10

dtype: int64

Master Missing Values:

```
player_id      0
player_name    0
season         0
position       0
nfl_team       0
draft_round    0
draft_pick_overall 0
age_on_draft_day 0
height_inches  0
weight_lbs     0
college        0
college_games_played 0
career_yards   0
career_touchdowns 0
rookie_total_ppr_points 10
season_positional_finish 10
fantasy_relevant_binary 10
dtype: int64
```

In [23]: *# Check data types*

```
master_df.info()
```

```

<class 'pandas.DataFrame'>
RangeIndex: 164 entries, 0 to 163
Data columns (total 17 columns):
 #   Column                Non-Null Count  Dtype
---  -
 0   player_id             164 non-null    str
 1   player_name           164 non-null    str
 2   season                164 non-null    int64
 3   position              164 non-null    str
 4   nfl_team              164 non-null    str
 5   draft_round           164 non-null    int64
 6   draft_pick_overall    164 non-null    int64
 7   age_on_draft_day      164 non-null    int64
 8   height_inches         164 non-null    int64
 9   weight_lbs            164 non-null    int64
10   college               164 non-null    str
11   college_games_played  164 non-null    int64
12   career_yards          164 non-null    int64
13   career_touchdowns     164 non-null    int64
14   rookie_total_ppr_points 154 non-null    float64
15   season_positional_finish 154 non-null    float64
16   fantasy_relevant_binary 154 non-null    float64
dtypes: float64(3), int64(9), str(5)
memory usage: 21.9 KB

```

In [24]: *# Check Season Count*

```
master_df["season"].value_counts().sort_index()
```

```
Out[24]: season
2010      8
2011      8
2012     11
2013      5
2014      9
2015     10
2016      8
2017     11
2018     11
2019      8
2020     11
2021     13
2022      7
2023     10
2024     14
2025     10
2026     10
Name: count, dtype: int64
```

```
In [25]: # Create Train/Test/Predict Split

train_df = master_df[
    (master_df["season"] >= 2010) &
    (master_df["season"] <= 2022)
].copy()

test_df = master_df[
    (master_df["season"] >= 2023) &
    (master_df["season"] <= 2025)
].copy()

predict_2026_df = master_df[
    master_df["season"] == 2026
].copy()

print("Train:", train_df.shape)
print("Test:", test_df.shape)
print("Predict 2026:", predict_2026_df.shape)
```

Train: (120, 17)
Test: (34, 17)
Predict 2026: (10, 17)

In [26]: *# Check Target Columns*

```
target_regression = "rookie_total_ppr_points"
target_classification = "fantasy_relevant_binary"

print("Missing regression targets:")
print("Train:", train_df[target_regression].isnull().sum())
print("Test:", test_df[target_regression].isnull().sum())
print("Predict 2026:", predict_2026_df[target_regression].isnull().sum())

print("\nMissing classification targets:")
print("Train:", train_df[target_classification].isnull().sum())
print("Test:", test_df[target_classification].isnull().sum())
print("Predict 2026:", predict_2026_df[target_classification].isnull().sum())
```

Missing regression targets:

Train: 0

Test: 0

Predict 2026: 10

Missing classification targets:

Train: 0

Test: 0

Predict 2026: 10

In [27]: *# Save Split Datasets*

```
train_df.to_csv("data/processed/train_2010_2022.csv", index=False)
test_df.to_csv("data/processed/test_2023_2025.csv", index=False)
predict_2026_df.to_csv("data/processed/predict_2026.csv", index=False)

print("Files saved successfully.")
```

Files saved successfully.